Environmental effects on Spatially-resolved Star Formation and Interstellar Medium Properties in the Virgo Cluster

Description

Galaxy evolution is predominantly governed by the balance between gas supply, star formation, and metal enrichment; while over-dense environments can perturb this balance, impacting the galaxy evolution process. Leveraging the state-of-the-art MAUVE-MUSE and MAUVE-ALMA observations of 40 Virgo late-type galaxies at ~100 pc resolution, this project will test whether a spatially-resolved fundamental metallicity relation (rFMR) exists and how it varies with environment. First, we will map the rFMR surface across all available galaxies' inner discs to study its existence and quantify cluster-driven offsets. Second, we will incorporate molecular and atomic gas to evaluate whether gas content explains the scatter in the resolved Star Forming Main Sequence and yields a tighter rFMR plane when substituted for gas-phase metallicity. Additionally, we will explore a stellar rFMR by replacing gas-phase with stellar metallicity to connect present enrichment to past star-formation histories. Together, these steps deliver an environmentally aware, spatially-resolved view of how gas, stars, and metals co-regulate galaxy evolution in a dense cluster environment.

- Project

Project / Activity overview

RDMP ID
P002431

ROAP Grant Application ID (if applicable)

Short Project/Activity name (maximum 8 characters)
22402136

UWA School/Centre/Institute
International Centre for Radio Astronomy Research - ICRAR

Start date
04/07/2025

End date
09/30/2028

Funding source

University of Western Australia

Is this a Higher Degree Research (HDR) project?
Yes

FoR Codes
51 - PHYSICAL SCIENCES

SEO Codes
28 - EXPANDING KNOWLEDGE

- People

People

UWA Chief Investigator (CI)

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HDR candidate (when they are not the CI)

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UWA Investigators

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Non-UWA Investigators

Name	Email	ORCID

RDMP administrators

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- Data description

Data description

Description of data collection

MAUVE

Type of data

Digital data

Are you doing research with any of the sanctioned countries listed on the Australian Government's Department of Foreign Affairs and Trade (DFAT) website?

No

Who will own the copyright and intellectual property of the research data?

Unknown or still to be decided

Data sensitivities

Type of sensitivities:

Public

Please select if your data is classified as one of the following:

- No sensitive content
- Able to published unaltered immediately
- Anonymised sensitive data

Data Classification Type

Public

Type of highly restricted data

- Data storage

Digital data storage

UWA research data storage options

New MS Teams/Sharepoint Research Site - recommended to use for collaboration and storage of project documents

Existing storage

IRDS storage volume

10TB

Reason for 10TB

Please describe your exemption justification including why it can't be stored on UWA storage and where is the store located?

Storage name for new location

RES-ICRAR-22402136-P002431

Data storage compliance statements

UWA-managed storage platforms meet standard security, backup, compliance and management requirements for research funders and ethics approval.

The full compliance statements are included in the PDF download of a saved RDMP, or they can be viewed in the Research Data Management Toolkit.

- Ethics

Ethics

Is ethics approval required?

No

Access after the project

-

For projects storing data in Microsoft Teams/Sharepoint:

The data generated in the proposed research project will be stored according to UWA's Research Integrity Policy.

This policy was developed to ensure that UWA research aligns with the principles and practices of the Australian Code for the Responsible Conduct of Research including those relating to research data storage.

The research data for the proposed project will be stored in UWA provisioned Microsoft Teams/Sharepoint service. This is located on Australian servers, is backed up, includes disaster recovery provisions, and can store data classified up to Highly Restricted. All stores are monitored for unauthorised and or malicious access, use and disclosure.

Data is accessible only to project collaborators identified by the project owner and access requires multifactor authentication using institutional usernames and passwords.

Data stored in Microsoft Teams/SharePoint is subject to the Western Australia University Sector Disposal Authority (WAUSDA) for retention and disposal.

- Upload attachments

Upload attachments

- Data Storage Requests

Data Storage Requests

Data storage request type	Storage path	Stage